

Ribonuclease A (RNase A) is a well-studied secreted monomeric enzyme involved in RNA degradation. Specifically, RNase A is an endonuclease that acts in the middle of an RNA strand by hydrolyzing the phosphodiester bond on the 3' side of a pyrimidine nucleotide (Figure 1).

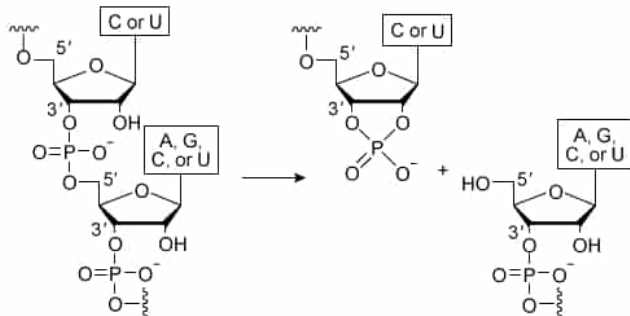


Figure 1 Reaction catalyzed by RNase A

Researchers studying the stability of RNase A first purified the enzyme from bovine pancreas and then fully denatured the enzyme with 8 M aqueous urea under reducing conditions, leaving only the amino acid sequence intact. Under these conditions, the researchers found that the enzyme lost all measurable activity. The researchers then gradually removed the denaturants by dialysis. Dialyzed enzyme was treated with a mild oxidizing agent before undergoing another activity assay. The researchers found that approximately 90% of activity was restored.

In a follow-up experiment, researchers treated the denatured enzyme with oxidizing agents first, prior to removing the urea by dialysis. Under these conditions, only approximately 2% of activity was recovered.

Based on the passage, the functional RNase A recovered after the removal of denaturants and subsequent treatment with oxidizing agent displays which levels of protein structure?

- I. Primary
- II. Secondary
- III. Tertiary
- IV. Quaternary

☐ A. III only [7%]

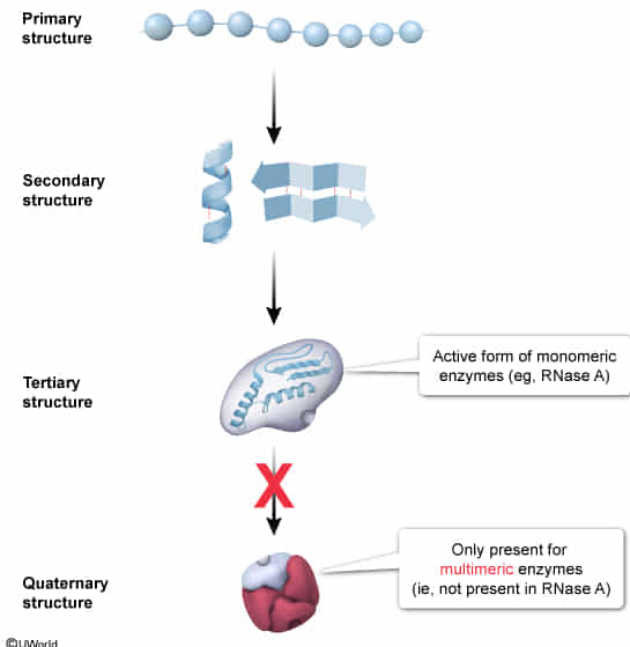
- ✓ ☒ B. I, II, and III only [58%]
☐ C. II, III, and IV only [17%]
☐ D. I, II, III, and IV [16%]

Correct

58% answered correctly

Explanation:

Quaternary structure is only present for multimeric proteins



Most enzymes, including RNase A, are catalytically active proteins. The activity of an enzyme, as with all proteins, is strongly linked to its **structure**. To be active, a protein must be folded in its correct structure, also called its **native conformation**. In contrast, misfolded, inactive proteins are considered **denatured**.

Protein structure can be divided into four levels:

1. **Primary structure** is the chemical structure of proteins, which is stabilized by **covalent bonds**. Because proteins are **polymers of amino acids**, each of which has a defined **chemical structure**, the primary structure can also be described by the protein's **amino acid sequence** (also called the protein's **primary sequence**), which is stabilized by **peptide bonds**. The passage states that the amino acid sequence **remains intact** throughout the experiment, and therefore the functional protein has an intact primary structure. (Number I)
2. **Secondary structure** is the local arrangement of the **peptide backbone**,

which is stabilized by **hydrogen bonds** between backbone amide hydrogens and backbone carbonyl oxygens. The most common examples of secondary structure are α -helices and β -sheets. Because the enzyme is functional and therefore in its native conformation, it must have an **intact secondary structure**. (Number II)

3. **Tertiary structure** is the overall three-dimensional shape of a **single polypeptide chain**. Tertiary structure is driven by **hydrophobic interactions** and is further stabilized by other **noncovalent** interactions (eg,

hydrogen bonds, electrostatic interactions) involving the **side chains** of the amino acids within the single polypeptide. Because the enzyme is functional and therefore in its native conformation, it must have an **intact tertiary structure**. (Number III)

4. **Quaternary structure** is the result of **intermolecular interactions** between **multiple polypeptides**, which come together to form a single functional protein. Quaternary structure is stabilized by forces similar to those that stabilize tertiary structure (ie, hydrophobic interactions, hydrogen bonds, electrostatic interactions).

(Number IV) Quaternary structure **exists only for multimeric proteins**. The passage states that RNase A is a **monomeric protein**. Therefore, functional RNase A enzymes do *not* consist of the multiple polypeptide chains required for quaternary structure.

Educational objective:

Proteins and enzymes require an intact structure to maintain their function. Protein structure can be divided into four levels. The fourth level, quaternary structure, exists only for multimeric proteins.

Time spent:0 sec

Subject:
Biochemistry

QID:403827

System:
Amino Acids and Proteins

Ribonuclease A (RNase A) is a well-studied secreted monomeric enzyme involved in RNA degradation. Specifically, RNase A is an endonuclease that acts in the middle of an RNA strand by hydrolyzing the phosphodiester bond on the 3' side of a pyrimidine nucleotide (Figure 1).

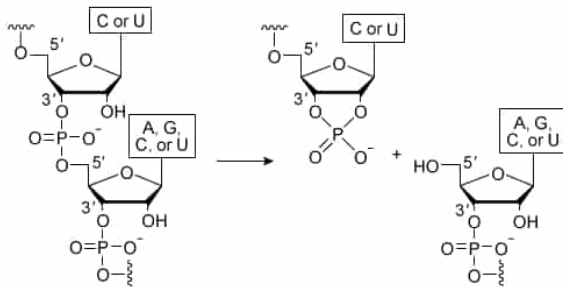


Figure 1 Reaction catalyzed by RNase A

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In a follow-up experiment, researchers treated the denatured enzyme with oxidizing agents first, prior to removing the urea by dialysis. Under these conditions, only approximately 2% of activity was recovered.

Which of the following conclusions about protein folding is best supported by the information in the passage?

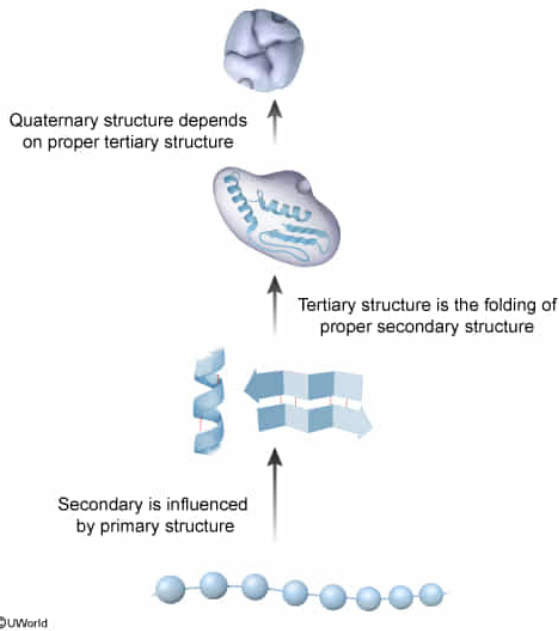
- ☐ A. Proteins are folded properly only when folded co-translationally. [11%]
- ☐ B. The purified RNase A sample was contaminated with chaperones that assisted in protein refolding. [20%]
- ✓ ☒ C. Higher levels of RNase A protein structure are encoded in the primary structure. [55%]
- ☐ D. All denatured proteins can spontaneously refold to a native conformation. [12%]

Correct

55% answered correctly

Explanation:

Higher levels of protein structure are encoded by lower levels of protein structure



Protein structure can be subdivided into **four levels**. Though each level can be individually described, each level of structure **depends** on a correct structure in the **lower levels** that precede it.

- **Quaternary structure** is the arrangement of **multiple polypeptides** forming a single multimeric unit. Quaternary structure requires each **individual subunit** to have its own **properly folded structure** (ie, tertiary structure).
- **Tertiary structure** is driven by the **hydrophobic collapse** of individual polypeptides, leading to a **hydrophobic interior protein core**. Although hydrophobic *side chains* are stable in this core, **polar peptide backbones** (arising from the amide hydrogen and the carbonyl oxygen) are not, unless the polar functional groups are stabilized by the **hydrogen bonds** of secondary structure.
- **Secondary structure** similarly depends on the primary structure. Although secondary structure hydrogen bonds are between **backbone functional groups**, side chain identities can affect secondary structure stability. For example, flexible **glycine residues** or rigid **proline residues** can both introduce kinks to disrupt α -helices. Similarly, β -sheets can be **stabilized** or destabilized by side chain interactions between strands.

The passage states that RNase A was fully **denatured** until only the amino acid sequence (ie, **primary structure**) remained intact. Nevertheless, RNase A was able to refold to a functional state (ie, regain higher levels of structure) without an external energy source. Based on this evidence, it can be concluded that the **higher levels of RNase A structure are encoded by its primary structure** due to the dependency of protein structure levels.

(Choice A) Although most proteins do fold co-translationally *in vivo*, the renaturation experiment in the passage shows that proteins can still fold properly long after translation

is complete.

(Choice B) **Chaperones** are proteins that assist in folding in an ATP-dependent manner; however, no ATP was added to facilitate refolding in the renaturation experiment. Moreover, as chaperones are proteins themselves, they would likely have been denatured and rendered inactive during the denaturation by urea.

(Choice D) Although the passage describes the renaturation of a single protein (RNase A), the results for a single protein cannot be generalized to all proteins without additional evidence.

Educational objective:

Protein structure is subdivided into four levels. The three higher levels of protein structure are dependent on the proper configuration of the lower levels of protein structure.

Time spent:0 sec

Subject:
Biochemistry

QID:403828

System:
Amino Acids and Proteins

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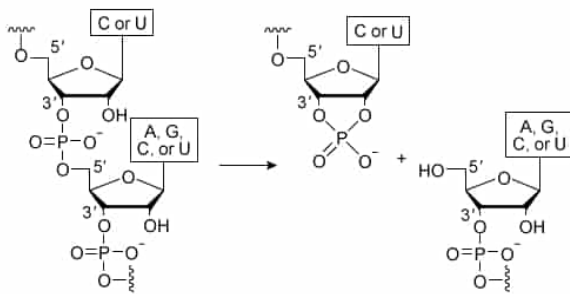


Figure 1 Reaction catalyzed by RNase A

Researchers studying the stability of RNase A first purified the enzyme from bovine pancreas and then fully denatured the enzyme with 8 M aqueous urea under reducing conditions, leaving only the amino acid sequence intact. Under these conditions, the researchers found that the enzyme lost all measurable activity. The researchers then gradually removed the denaturants by dialysis. Dialyzed enzyme was treated with a mild oxidizing agent before undergoing another activity assay. The researchers found that approximately 90% of activity was restored.

In a follow-up experiment, researchers treated the denatured enzyme with oxidizing agents first, prior to removing the urea by dialysis. Under these conditions, only approximately 2% of activity was recovered.

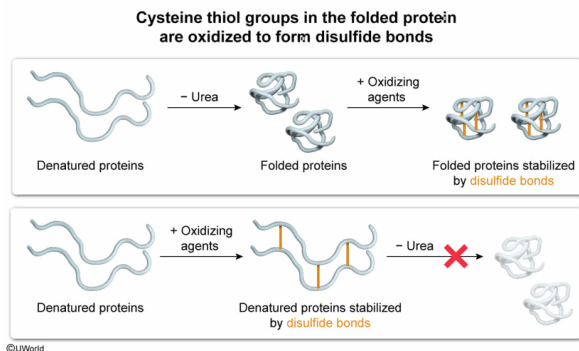
Why was a mild oxidizing agent added during the experiments?

- ☒ A. To stabilize folded proteins by formation of disulfide bonds [36%]
- ☐ B. To stimulate denatured proteins to form disulfide bonds so that they can fold [54%]
- ☐ C. To stabilize folded protein by breaking disulfide bonds [2%]
- ☐ D. To stimulate denatured proteins to break disulfide bonds so that they can fold [6%]

Correct

35% answered correctly

Explanation:



After the initial **hydrophobic collapse** of a polypeptide, additional **noncovalent forces** act to partially stabilize a protein's tertiary structure. However, these noncovalent forces can continue to break and reform even at physiological temperature. The tertiary structure of a protein can therefore be stabilized even further by the addition of covalent **disulfide bonds**.

Disulfide bonds form when the **thiol** ($-\text{SH}$) group of a **cysteine** residue reacts with another nearby **cysteine**. The resulting covalent linkage ($-\text{S}-\text{S}-$) is referred to as a **disulfide bond** or a **disulfide bridge** and strongly links regions of protein together. If the protein was folded correctly beforehand, the protein is **stabilized** to resist denaturation. However, if the protein was folded incorrectly before formation of disulfide bonds, a cysteine may form a disulfide bond to an **incorrect cysteine partner** and stabilize a *denatured* conformation.

The bond between the sulfur and the hydrogen of a free thiol is a **polar bond**. Sulfur is more **electronegative** than hydrogen, and therefore it has a relatively **high electron density** around it. In contrast, because a disulfide bond is between two equally electronegative sulfur atoms, this bond is **nonpolar**. Each sulfur **loses some electron density** as electrons are equally shared between them. This **loss of electron density** is **oxidation**. Therefore, **oxidizing agents** were added to **oxidize free thiols to form disulfide bonds**, which stabilizes the conformation of the protein.

(Choice B) Disulfide bonds stabilize proteins in the conformation in which they are oxidized. The passage indicates that if oxidation happens *prior* to removing urea, RNase A remains denatured and nonfunctional.

(Choices C and D) Oxidizing agents are used to *form* disulfide bonds, not break them.

Educational objective:

Disulfide bonds ($-\text{S}-\text{S}-$) are formed by the oxidation of two nearby thiol ($-\text{SH}$) groups. Disulfide bonds form a covalent linkage that strongly stabilizes local protein structure.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403829
System:
Amino Acids and Proteins

Ribonuclease A (RNase A) is a well-studied secreted monomeric enzyme involved in RNA degradation. Specifically, RNase A is an endonuclease that acts in the middle of an RNA strand by hydrolyzing the phosphodiester bond on the 3' side of a pyrimidine nucleotide (Figure 1).

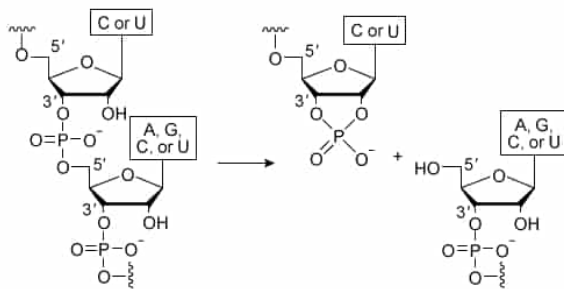


Figure 1 Reaction catalyzed by RNase A

Researchers studying the stability of RNase A first purified the enzyme from bovine pancreas and then fully denatured the enzyme with 8 M aqueous urea under reducing conditions, leaving only the amino acid sequence intact. Under these conditions, the researchers found that the enzyme lost all measurable activity. The researchers then gradually removed the denaturants by dialysis. Dialyzed enzyme was treated with a mild oxidizing agent before undergoing another activity assay. The researchers found that approximately 90% of activity was restored.

In a follow-up experiment, researchers treated the denatured enzyme with oxidizing agents first, prior to removing the urea by dialysis. Under these conditions, only approximately 2% of activity was recovered.

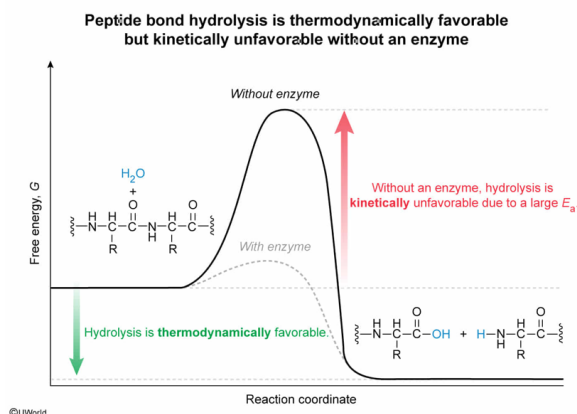
The peptide bonds maintaining the amino acid sequence were left intact in aqueous solution, despite the presence of denaturants. This is because the hydrolysis of peptide bonds:

- ☒ A. is kinetically slow despite being thermodynamically spontaneous. [28%]
- ☐ B. is thermodynamically nonspontaneous despite being kinetically fast. [31%]
- ☐ C. requires water molecules, which were not supplied. [26%]
- ☐ D. requires energy input from GTP, which was not supplied. [13%]

Correct

27% answered correctly

Explanation:



The **spontaneity** and **rate** of a chemical reaction are described by its **thermodynamics** and its **kinetics**, respectively. Spontaneous reactions have a **negative free energy change** ($-\Delta G$) and tend to proceed in the **forward direction** without the need for an external energy source (eg, ATP). The **hydrolysis of macromolecules** (eg, proteins, polysaccharides, nucleic acids) is generally a spontaneous reaction.

However, a reaction that is thermodynamically favorable (and therefore **spontaneous**) may not proceed at an appreciable rate if it is **kinetically unfavorable** (ie, slow). For example, proteins, polysaccharides, and nucleic acids have long half-lives in living cells unless and until the cell requires them to be broken down. This is because **hydrolysis of these molecules is kinetically slow** due to a large **activation energy**, which prevents the reaction from proceeding at the mild temperatures within living organisms. When hydrolysis is needed, organisms activate enzymes, which allow the reactions to proceed by **lowering the activation energy**.

Because peptide bond hydrolysis does not occur without enzymes despite being thermodynamically favorable (ie, has a $-\Delta G$), it is **kinetically slow but thermodynamically spontaneous**.

(Choices B and D) Peptide bond *formation* is nonspontaneous and requires energy from the nucleoside triphosphate GTP to occur at the ribosome. In contrast, peptide bond *hydrolysis* is spontaneous and does not require GTP.

(Choice C) The passage states that the denaturation occurred in an aqueous solution. Therefore, excess amounts of water were available.

Educational objective:

The bonds that form macromolecules (eg, peptide bonds in proteins, glycosidic linkages in carbohydrates, phosphodiester bonds in nucleic acids) require energy input to form. In contrast, the hydrolysis reactions that break them are thermodynamically favorable. However, these reactions are also kinetically unfavorable without an enzyme, allowing the macromolecules to persist with long half-lives once formed.

Time spent: 0 sec

Subject:

Biochemistry

QID: 403830

System:

Amino Acids and Proteins

Apolipoprotein L-I (APOL1) is an important protein of primate innate immunity, particularly against trypanosome infection. APOL1 is a 43 kDa protein that is secreted mainly from hepatocytes as a component of certain high-density lipoprotein (HDL) particles that also contain targeting proteins that bind to trypanosomes and induce endocytosis of the HDL particle.

Acidification of the trypanosomal endosome induces APOL1 to transfer from the aqueous lumen to integrate into the endosomal membrane, forming a closed ion channel. As the endosomal membrane is recycled back into the plasma membrane, the pH neutralization of the *cis* side (ie, luminal side in endosomes, extracellular side on the plasma membrane) then activates the ion channel, allowing sodium and potassium cations to flow through. In other words, APOL1 acts as a ligand-gated ion channel that opens when its ligand (H⁺) unbinds. Depolarization and dysregulation of ion flux then leads to osmotic lysis of the trypanosome parasite.

To structurally characterize the topology of the APOL1 ion channel, researchers used bioinformatics methods based on the APOL1 primary sequence to predict transmembrane regions as well as an experimental technique called the substituted cysteine accessibility method (SCAM) to determine extracellular and intracellular localization of the intervening loops. To perform SCAM, the researchers substituted specific amino acids in these loops with cysteine residues and allowed the APOL1 mutants to form functional ion channels. A 53 kDa thiol-reactive reagent (*N*-(3-maleimidylpropionyl)biocytin-streptavidin, MPBS) was then added to the *cis* face (ie, extracellular side) of the channels, and the molecular weight of the APOL1-MPBS complex was analyzed by nonreducing SDS-PAGE followed by western blot. This experiment was repeated using MPB without streptavidin (530 Da) and with MPBS along with a small amount of the gentle detergent n-dodecyl-β-D-maltoside (DDM). The results are shown in Figure 1.

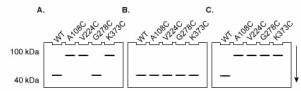


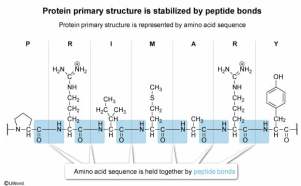
Figure 1 Western blots of APOL1 reacted with (A) MPBS, (B) MPB without streptavidin, and (C) MPBS and 1% w/v DDM.

The bioinformatic study described in the passage relies on the level of protein structure stabilized by:

- I. electrostatic interactions.
 - II. hydrogen bonds.
 - III. hydrophobic interactions.
 - IV. peptide bonds.
- ☐ A. II only [5%]
☒ B. IV only [53%]
☐ C. III and IV only [14%]
☐ D. I, II, III, and IV [26%]

Correct
52% answered correctly

Explanation:



Proteins are **biological macromolecules** that serve various roles in living systems. The activity and function of a protein depends heavily on its **structure**. Proteins exhibit **four levels of structure**.

- **Primary structure** is the chemical structure of proteins and is stabilized by **covalent bonds**. Because proteins are **polymers of amino acids**, each of which has a defined **chemical structure**, the primary structure can also be described by the protein's **amino acid sequence** (also called the protein's **primary sequence**). Amino acids within a protein polymer are connected to adjacent amino acids by covalent **peptide bonds**; therefore, primary structure is **stabilized by peptide bonds**.
- **Secondary structure** is the local arrangement of the peptide backbone and is stabilized by **hydrogen bonds** between backbone amide hydrogens and backbone carbonyl oxygens. The most common examples of secondary structure are alpha helices and beta-pleated sheets.
- **Tertiary structure** is the overall three-dimensional shape of a **single polypeptide chain**. Tertiary structure is driven by **hydrophobic interactions** and is further stabilized by other **noncovalent** interactions (eg, hydrogen bonds, electrostatic interactions) involving the **side chains** of the amino acids within the single polypeptide.
- **Quaternary structure** is the **intermolecular interactions** that result from **multiple polypeptides** coming together to form a single functional protein. Quaternary structure is stabilized by similar forces as tertiary structure (ie, hydrophobic interactions, hydrogen bonds, electrostatic interactions).

The passage states that bioinformatics methods based on the APOL1 **primary sequence** were used to predict transmembrane regions. The primary sequence refers to the protein's **primary structure**, which is stabilized by covalent bonds such as the **peptide bonds** between individual amino acids (**Number IV**).

(Numbers I, II, and III) Hydrogen bonds between amino acid backbone groups stabilize a protein's secondary structure; hydrogen bonds, electrostatic interactions, and hydrophobic interactions between side chains all stabilize a protein's tertiary and quaternary structure. However, the bioinformatics study described used only the protein's **primary structure** to predict transmembrane regions.

Educational objective:

Protein structure can be subdivided into four levels. The first level, primary structure, can be described by the protein's amino acid sequence, which is stabilized by the covalent peptide bonds that link adjacent amino acids within the sequence.

Apolipoprotein L-I (APOL1) is an important protein of primate innate immunity, particularly against trypanosome infection. APOL1 is a 43 kDa protein that is secreted mainly from hepatocytes as a component of certain high-density lipoprotein (HDL) particles that also contain targeting proteins that bind to trypanosomes and induce endocytosis of the HDL particle.

Acidification of the trypanosomal endosome induces APOL1 to transfer from the aqueous lumen to integrate into the endosomal membrane, forming a closed ion channel. As the endosomal membrane is recycled back into the plasma membrane, the pH neutralization of the *cis* side (ie, luminal side in endosomes, extracellular side on the plasma membrane) then activates the ion channel, allowing sodium and potassium cations to flow through. In other words, APOL1 acts as a ligand-gated ion channel that opens when its ligand (H^+) unbinds. Depolarization and dysregulation of ion flux then leads to osmotic lysis of the trypanosome parasite.

To structurally characterize the topology of the APOL1 ion channel, researchers used bioinformatics methods based on the APOL1 primary sequence to predict transmembrane regions as well as an experimental technique called the substituted cysteine accessibility method (SCAM) to determine extracellular and intracellular localization of the intervening loops. To perform SCAM, the researchers substituted specific amino acids in these loops with cysteine residues and allowed the APOL1 mutants to form functional ion channels. A 53 kDa thiol-reactive reagent (*N*-(3-maleimidylpropionyl)biocytin–streptavidin, MPBS) was then added to the *cis* face (ie, extracellular side) of the channels, and the molecular weight of the APOL1-MPBS complex was analyzed by nonreducing SDS-PAGE followed by western blot. This experiment was repeated using MPB without streptavidin (530 Da) and with MPBS along with a small amount of the gentle detergent n-dodecyl- β -D-maltoside (DDM). The results are shown in Figure 1.



Figure 1 Western blots of APOL1 reacted with (A) MPBS, (B) MPB without streptavidin, and (C) MPBS and 1% w/v DDM.

Based on the passage, the inclusion of which amino acid is most likely responsible for the changing localization (soluble luminal to integral membrane) of APOL1 upon pH manipulation?

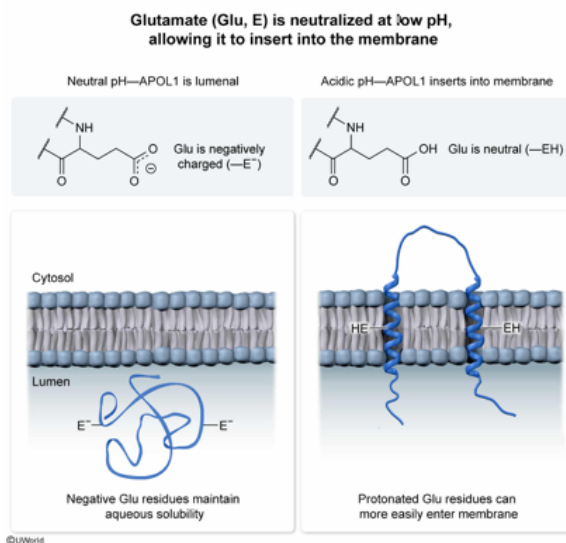
- ☐ A. Isoleucine [26%]
- ✓ ☒ B. Glutamate [52%]
- ☐ C. Histidine [8%]

☐ D. Lysine [12%]

Correct

53% answered correctly

Explanation:



The properties of an amino acid's **side chain** dictate the structural and functional role that the amino acid plays in the context of the larger protein. For example, **acidic** or **basic** side chains respond to pH changes or participate in acid-base catalytic mechanisms, **hydrophilic** side chains maintain solubility in aqueous solutions, and **hydrophobic** side chains drive hydrophobic interactions to form the protein's hydrophobic core or a transmembrane segment.

The passage states that APOL1 starts off in the aqueous lumen but partitions to the membrane as the lumen acidifies. Thus, the amino acids responsible for this shift must be **responsive to pH changes**. Furthermore, the amino acids should be **soluble** in the aqueous lumen at a **neutral pH** but **less soluble when acidified**.

Glutamate is an acidic amino acid that changes protonation state at different pH levels. At neutral pH, the glutamate side chain is **negatively charged**, and therefore **highly soluble**. When the pH drops, glutamate becomes protonated, which neutralizes its charge and makes it **less water-soluble under acidic conditions**. Therefore, the inclusion of **glutamate** is most likely responsible for the changing localization of APOL1.

(Choice A) **Isoleucine** has a hydrophobic side chain at all pH values. Although its inclusion would assist in partitioning the membrane at low pH, it would not be stable in the aqueous lumen at neutral pH.

(Choice C) **Histidine** is a basic amino acid with pK_a around 6.0. The histidine side chain is mostly neutral (ie, less water-soluble) at neutral pH and is positively charged (ie, more water-soluble) under acidic conditions. These properties are the *opposite* of what is expected for APOL1.

(Choice D) Lysine is a basic amino acid that is positively charged under both neutral and acidic conditions. The inclusion of positively charged lysine would be unlikely to permit the integration into the lipid bilayer membrane.

Educational objective:

Acidic and basic amino acids change their charges at different pH values. Although negatively charged and highly hydrophilic at neutral pH, acidic amino acids like glutamate can be protonated at acidic pH to become more lipid-soluble.

Time spent:0 sec

Subject:
Biochemistry

QID:403832

System:
Amino Acids and Proteins

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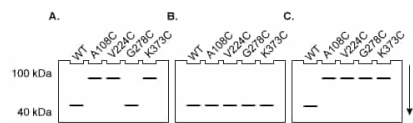


Figure 1 Western blots of APOL1 reacted with (A) MPBS, (B) MPB without streptavidin, and (C) MPBS and 1% w/v DDM.

Based on the data presented in the passage, which of the following best describes the membrane protein topology of the APOL1 channel?

- ☐ A.

Residue	Location
A108	Cytosolic
V224	Lumenal/Extracellular
G278	Cytosolic
K373	Lumenal/Extracellular

[4%]
- ☐ B.

Residue	Location
A108	Cytosolic
V224	Cytosolic
G278	Lumenal/Extracellular
K373	Cytosolic

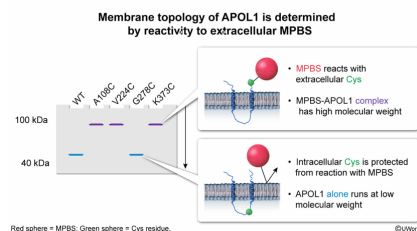
[40%]
- ☐ C.

Residue	Location
A108	Lumenal/Extracellular
V224	Cytosolic
G278	Lumenal/Extracellular
K373	Cytosolic

[6%]
- ☒ D.

Residue	Location
A108	Lumenal/Extracellular
V224	Lumenal/Extracellular
G278	Cytosolic
K373	Lumenal/Extracellular

[48%]

Explanation:

The substituted cysteine accessibility method (SCAM) technique uses a combination of cysteine mutations to APOL1 and reactivity of these mutants to MPBS to determine the **membrane topology** (ie, which protein regions are intracellular, in the membrane, or extracellular/luminal) of APOL1. A single cysteine mutation is added to APOL1, the resulting mutant is induced to traffic into the membrane and then the large reagent MPBS is added to the extracellular side of the membrane, where it can react with available cysteines.

Because of the large size of MPBS (53 kDa), it is unable to pass through the **cellular membrane**. Therefore, MPBS will **only react with extracellular cysteines**. Upon reaction, MPBS forms a covalent bond to APOL1, yielding a **96 kDa complex** (53 kDa + 43 kDa) whose molecular weight can be assessed by SDS-PAGE followed by a western blot. The APOL1 mutants A108C, V224C, and K373C all produced a band at 96 kDa, indicating that they reacted with MPBS and that those positions (**A108, V224, and K373**) are **extracellular**.

In contrast, **intracellular cysteines do not react with MPBS**. MPBS does not cross the membrane, and these cysteines do not react to form a complex. Unreacted APOL1 maintains its wild-type molecular weight of 43 kDa. The mutant G278C produced a band at 43 kDa, indicating that the position **G278 is intracellular**.

(Choices A and C) These tables show an alternating pattern of extracellular and cytosolic localization, which would only be expected if exactly one location were probed from each loop between each transmembrane region. If a loop was skipped or if a loop were probed multiple times, it may produce a non-alternating pattern. The western blot suggests that this is the case.

(Choice B) If G278 were extracellular and the other positions were intracellular, then the G278C mutant would have been expected to shift molecular weight and the other positions to remain at the wild-type weight. However, the opposite result is displayed in Figure 1A.

Educational objective:

The plasma membrane is an effective barrier that prevents the passage of large molecules. With an intact membrane, large reagents only react with the portions of the membrane protein on the same side of the membrane as the reagent. The reactivity of reagents with specific amino acids of a membrane protein can be used to determine extracellular and intracellular localization of protein loops.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403833
System:
Amino Acids and Proteins

Apolipoprotein L-I (APOL1) is an important protein of primate innate immunity, particularly against trypanosome infection. APOL1 is a 43 kDa protein that is secreted mainly from hepatocytes as a component of certain high-density lipoprotein (HDL) particles that also contain targeting proteins that bind to trypanosomes and induce endocytosis of the HDL particle.

Acidification of the trypanosomal endosome induces APOL1 to transfer from the aqueous lumen to integrate into the endosomal membrane, forming a closed ion channel. As the endosomal membrane is recycled back into the plasma membrane, the pH neutralization of the *cis* side (ie, luminal side in endosomes, extracellular side on the plasma membrane) then activates the ion channel, allowing sodium and potassium cations to flow through. In other words, APOL1 acts as a ligand-gated ion channel that opens when its ligand (H^+) unbinds. Depolarization and dysregulation of ion flux then leads to osmotic lysis of the trypanosome parasite.

To structurally characterize the topology of the APOL1 ion channel, researchers used bioinformatics methods based on the APOL1 primary sequence to predict transmembrane regions as well as an experimental technique called the substituted cysteine accessibility method (SCAM) to determine extracellular and intracellular localization of the intervening loops. To perform SCAM, the researchers substituted specific amino acids in these loops with cysteine residues and allowed the APOL1 mutants to form functional ion channels. A 53 kDa thiol-reactive reagent (*N*-(3-maleimidylpropionyl)biocytin-streptavidin, MPBS) was then added to the *cis* face (ie, extracellular side) of the channels, and the molecular weight of the APOL1-MPBS complex was analyzed by nonreducing SDS-PAGE followed by western blot. This experiment was repeated using MPB without streptavidin (530 Da) and with MPBS along with a small amount of the gentle detergent n-dodecyl- β -D-maltoside (DDM). The results are shown in Figure 1.

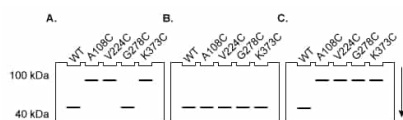


Figure 1 Western blots of APOL1 reacted with (A) MPBS, (B) MPB without streptavidin, and (C) MPBS and 1% w/v DDM.

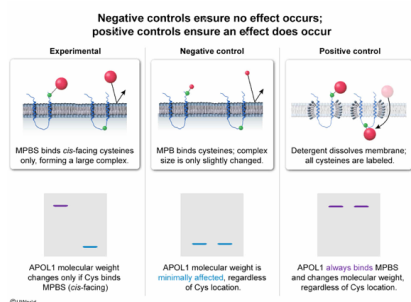
What were the purposes of the experiment with MPB without streptavidin and of the experiment with MPBS and detergent?

- ☐ A. They served as a positive and negative control, respectively [26%]
- ☒ B. They served as a negative and positive control, respectively [65%]
- ☐ C. Both served as positive controls [4%]
- ☐ D. Both served as negative controls [3%]

Correct

66% answered correctly

Explanation:



Controls are vital components of experimental design that help ensure that the investigator interprets results correctly.

Negative controls are experimental conditions in which an experimental change is **not expected** to be seen. Alternatively, **positive controls** are conditions in which an experimental change is **always expected** to be observed.

The passage describes a substituted cysteine accessibility method (SCAM) experiment, which involves the application of a large (53 kDa) reagent applied to one side of the membrane. Due to the large size of the reagent, it is **not expected to cross an intact membrane**, and can only react with extracellular-facing regions of APOL1. If the two react, this experimental change would be seen as a **large size change** on a western blot.

MPB without streptavidin is only 530 Da, approximately 1% of the size of APOL1. It is **not expected** to cause a significant size change on a western blot, regardless of whether it reacts with APOL1; therefore, **MPB without streptavidin serves as a negative control**. Negative controls help rule out the possibility that the cysteine mutations themselves (or some other confounding variable) cause a size change (ie, false positive).

Detergents disrupt the stability of the cell membrane and allow **MPBS** to cross through and react with both extracellular *and* intracellular cysteines. Consequently, a size change **is always expected** regardless of cellular localization; therefore, **MPBS with detergent acts as a positive control**. This positive control helps prevent false negatives by ensuring that intracellular cysteines *are* reactive, and that a negative result is only due to localization.

(Choice A) MPB is a *negative* control, and MPB with the detergent is the *positive* control.

(Choice C) MPB without streptavidin is expected to *not* cause a size change, and therefore acts as a *negative* control.

(Choice D) The inclusion of detergent is expected to allow MPBS to react with APOL1 *regardless of localization*; therefore, it acts as a *positive* control.

Educational objective:

Controls are critical components of experimental design. Negative controls are expected to produce no result and help rule out false positives. Positive controls are expected to produce results and help rule out false negatives.

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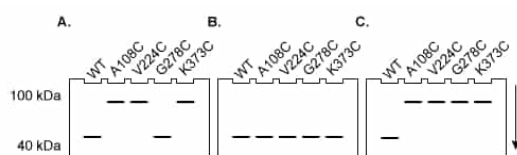


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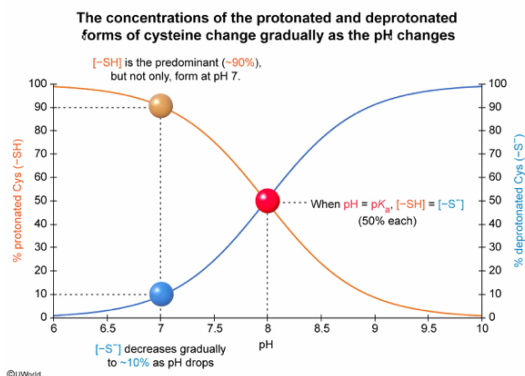
The cysteine-MPBS reaction is a redox reaction involving the nucleophilic attack on MPBS by a deprotonated cysteine side chain. If the reaction were attempted at pH 7.0, approximately what percent of cysteine side chains would be in the deprotonated, anionic form? (Note: Assume that the cysteine side chains all have pK_a of 8.0.)

- ☐ A. 0% [28%]
- ✓ ☒ B. 10% [51%]
- ☐ C. 50% [11%]
- ☐ D. 75% [8%]

Correct

50% answered correctly

Explanation:



The protonation state of **ionizable amino acids** is influenced by both the pH and the **pK_a value** of the side chain functional group. The question states that cysteine has a pK_a of 8.0; at pH values *above* 8.0, the predominant form is the deprotonated state, and at pH values *below* 8.0, the predominant form is the protonated state. A pH of 7 is below 8.0, indicating that the deprotonated form is *not* predominant; however, a significant proportion of cysteine residues are still deprotonated.

The protonated and deprotonated forms of a functional group form a conjugate acid–conjugate base system, also known as a **buffer system**. Given the pH and pK_a values for the deprotonation reaction, the relative concentration of the conjugate acid and conjugate base can be calculated using the **Henderson–Hasselbalch equation**:

$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

By substituting in the given values for pH and pK_a, the base:acid ratio is calculated as 1:10.

$$7.0 = 8.0 + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$-1.0 = \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$\frac{[\text{A}^-]}{[\text{HA}]} = 10^{-1.0}$$

$$\frac{[\text{A}^-]}{[\text{HA}]} = \frac{1}{10}$$

The percentage of cysteine in the deprotonated form (ie, base, A⁻) can be accurately **calculated** by dividing the numerator of this ratio (ie, base form of cysteine, 1) by the sum of the numerator plus the denominator (ie, total cysteine, 1 + 10 = 11). A less precise approximation can be quickly performed by rounding: $1/11 \approx 1/10 = \mathbf{10\%}$ of cysteines in the deprotonated, anionic form.

(Choice A) Although the deprotonated form is not the *predominant* form when $\text{pH} < \text{pK}_a$, a measurable percentage of cysteines in the deprotonated form still exists at a pH of 7.

(Choice C) Deprotonated cysteine would only be 50% of total cysteine if the pH equaled cysteine's pK_a value of 8.0.

(Choice D) A pH of 7.0 would be *below* the pK_a value of 8.0, and the *protonated* form would be the predominant species. If the protonated form were predominant, the deprotonated form would not make up over 50% of the total cysteine.

Educational objective:

The predominant protonation state of a functional group changes when the pH crosses the pK_a value. At pH values different from the pK_a , the concentrations of the predominant and minor forms can be calculated using the Henderson–Hasselbalch equation.

Time spent: 0 sec

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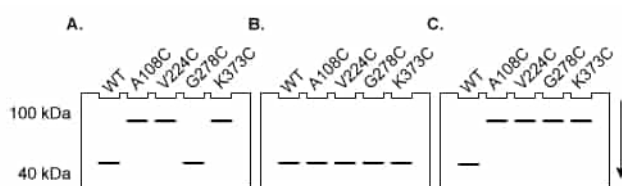


Figure 1 Western blots of APOL1 reacted with (A) MPBS, (B) MPB without streptavidin, and (C) MPBS and 1% w/v DDM.

Trypanosomes were transfected with a new ion channel, Channel X. Channel X

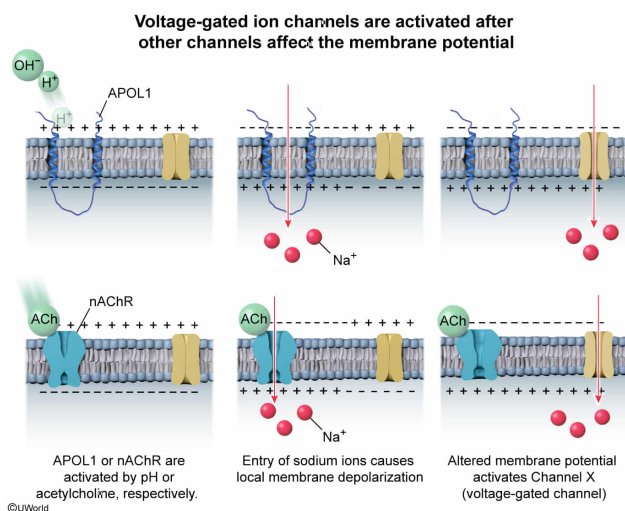
is found to open in trypanosomes that have been treated with APOL1 once pH neutralization occurs. Similarly, Channel X opens in trypanosomes that contain the nicotinic acetylcholine receptor (nAChR) when treated with acetylcholine (the ligand that opens nAChR). Neither pH nor acetylcholine open Channel X in trypanosomes *without* APOL1 or nAChR, respectively. Channel X is best described as a:

- ☐ A. mechanically gated ion channel. [12%]
- ☐ B. pH-activated channel. [3%]
- ☐ C. ligand-gated ion channel. [67%]
- ✓ ☒ D. voltage-gated ion channel. [16%]

Correct

15% answered correctly

Explanation:



Ion channels are membrane proteins that allow ions to cross the **semipermeable membrane**, traveling passively down their electrochemical gradient and altering the membrane potential. Ion channels can be constitutively (ie, always) open, or they can be **gated ion channels** (ie, only open in response to certain signals). Gated ion channels are broadly classified based on the type of stimulus to which they respond.

- **Ligand-gated ion channels** open and close based on physical interactions with a **ligand**. The nicotinic acetylcholine receptor (nAChR), natively found at the **neuromuscular junction**, opens upon binding the ligand acetylcholine and results in depolarization. pH-gated channels can be considered a subset of ligand-gated channels, where protons act as ligands. The APOL1 channel is a ligand-gated channel that activates upon the *removal*

of the proton ligand.

- **Mechanically gated ion channels** respond to mechanical forces like pressure or tension. These forces alter the dimensions of the fluid lipid bilayer membrane, and therefore cause conformational changes that result in the opening of the gate.
- **Voltage-gated ion channels** respond to **changes** in the membrane potential that have usually been caused by *other* channels. By responding to other ion channels such as APOL1 or nAChR, voltage-gated channels can **propagate** an electrical stimulus across the surface of a cell. Because Channel X only activates upon the opening of (and the subsequent resultant depolarization by) either APOL1 or nAChR, **Channel X is most likely a voltage-gated ion channel.**

(Choice A) The question makes no mention of a mechanical force being required to activate Channel X.

(Choice B) pH neutralization is stated to result in the activation of Channel X, *but only in the presence of APOL1*. The *depolarization* (ie, change in voltage) caused by APOL1 activation, rather than the pH itself, most likely activates Channel X.

(Choice C) The ligand acetylcholine is stated to result in the opening of Channel X *but only in the presence of nAChR*, which binds acetylcholine. The *depolarization* (ie, change in voltage) caused by nAChR activation, rather than the ligand itself, most likely activates Channel X.

Educational objective:

Voltage-gated ion channels activate upon a change in membrane potential. These changes in membrane potentials are typically first caused by other ion channels. In this way, voltage-gated ion channels propagate signals originated by other ion channels.

Time spent:0 sec

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